**ITA0610-Machine Learning**

**LAB EXPERIMENTS**

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## Lab 29: NumPy Statistics and Aggregation

### Aim

To compute descriptive statistics such as **mean, sum, standard deviation, minimum, and maximum** using NumPy aggregation functions.

### Algorithm

1. Import the NumPy library.
2. Create a two-dimensional NumPy array.
3. Calculate the mean of all elements.
4. Calculate the sum along rows and columns.
5. Calculate the standard deviation.
6. Find the minimum value for each row.
7. Find the maximum value in the array.
8. Display the results.

### Program

```
# Lab 29: NumPy Statistics and Aggregation


import numpy as np


# Create a 2D array

a = np.array([

    [1, 2, 3],

    [4, 5, 6]

])


# Mean

print("Mean:", a.mean())
```

```python
# Sum along columns
print("Sum along axis 0 (columns):", a.sum(axis=0))

# Sum along rows
print("Sum along axis 1 (rows):", a.sum(axis=1))

# Standard deviation
print("Std Dev:", a.std())

# Minimum value in each row
print("Min per row:", a.min(axis=1))

# Maximum value in the array
print("Max overall:", a.max())
```

**Sample Output**

```
Mean: 3.5
Sum along axis 0 (columns): [5 7 9]
Sum along axis 1 (rows): [ 6 15]
Std Dev: 1.707825127659933
Min per row: [1 4]
Max overall: 6
```

**Result**

Thus, **statistical aggregation functions were successfully performed on a NumPy array** using functions such as mean, sum, standard deviation, minimum, and maximum.